

Victor Kipkirui Korir

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About me

I am a geo-informatics professional who is passionate about mapping, Geographic Information Systems (GIS) analysis, location based services over the web and remote sensing image analysis. My core values are excellence, leadership, teamwork, honesty, integrity, fairness, professionalism and respect for diversity.

Skills

- **Software Skills:** Geographic Information Systems(GIS) in Qgis and Arcgis, Remote Sensing in various softwares such as SNAP, ERDAS and ENVI, Point Cloud processing, Git, Linux, Google Earth Engine(GEE), SEPAL, SQL.
- **Programming Languages:** R (rgee, rgdal, gstat, sf, terra, STAC), Python(numpy, geopandas, fiona, seaborn, scipy), OOP Java

Professional Experience

- **Consultative Group on International Agricultural Research(CGIAR) - Alliance of Bioversity and International Center for Tropical Agriculture (CIAT)**

Senior Research Associate-Data Science

July 2023 - Present

- Implemented data science analysis techniques, including machine learning, to explore the correlation between climate, conflict, and human security.
- Supported backend spatial analysis for the [Climate Security Observatory\(CSO\)](#), enhancing stakeholders' preparedness for climate security risks. My specific contributions can be viewed in the github repository.
- Engaged in collaborative efforts with government ministries and regional blocs to influence the development of both national and regional policies pertaining to climate change and human security. Significant accomplishments include in the formulation of the IGAD regional strategy on climate security and the Kenya National Climate Change Action Plan (NCCAP) 2023-2027.
- Actively participated in fundraising endeavors by assisting in the preparation of grant proposals and concept notes.

- **Center for Remote Sensing of Land Surfaces, University of Bonn**

Research Associate(Wissenschaftlicher Mitarbeiter)

March 2022 - July 2023

- Conducted comprehensive research to gather and organize information, summaries, and materials pertaining to Copernicus and its diverse components, including datasets and services.
- Established and expanded a network of contacts, encompassing potential Copernicus users across various user groups in multiple African nations.
- Evaluated user needs and requirements concerning Copernicus through methods such as interviews, surveys, and engagement at events like UN-Spider expert meetings, fostering dialogue with aforementioned contacts and user groups.
- Produced training resources, including online brochures, tutorials and video guides, covering various aspects of Copernicus applications. These products were made publicly available through: <https://www.zfl.uni-bonn.de/resources-1/training-module-handbook-contents/flood-module> and [UN-SPIDER knowledge portal](#).

- **Juelich Research Center (Forschungszentrum Jülich)**

Master Thesis Student

March 2022 - September 2022

- Conducted research and wrote a master thesis on land suitability analysis for placement of hydrogen Infrastructure using a multi-criteria decision making(MCDM) approach using a case study of ECOWAS nations in West Africa.

- **University of Bonn**

Student Research Assistant.

November 2020 - November 2021

- Performed systematic literature reviews on selected topics.

- Compiled teaching materials including module handbooks and presentation slides.
 - Assessed seasonal floods in Baringo county, Kenya using Sentinel 1 C-Band Synthetic Aperture Radar data (SAR) through Google Earth engine (GEE) in R using the RGEE interface.
 - Performed time series analysis of *Prosopis Juliflora* invasion in Marigat plains in Baringo county based on NDVI derived from Landsat images acquired from Google Earth Engine (GEE) using an R interface (RGEE).
- **agriBORA GmbH**
GIS and Remote Sensing Analyst. *December 2019 - September 2020*
- Researched on the state of the art in land use and plant classification. The focus was on the following crops: soybeans, rice, wheat, sorghum, maize, cassava.
 - Performed Land use and crop classification, using state-of-the-art GIS techniques and machine learning on freely available satellite images (e.g. Sentinel 2). Production of maps, metrics and readable reports.
 - Obtained training data for model calibration and validation This includes the data and locations of the cultivated crops.
- **Precision Agriculture for Development (PAD)**
GIS Consulting Intern *May 2019 - November 2019*
- Carried out a landscape analysis to determine the geo-spatial resources available for East Africa, with a focus on Kenya, but also Uganda, Rwanda and Ethiopia.
 - Built a database of geo-spatial layers for the countries of interest, focusing on agriculture (agro-ecological zones, land use, cultivation calendar), weather, climate and soils.
 - Integrated MODIS with GIS: production of interactive maps to illustrate local NDVI patterns.

Academic Qualification

- **Master of Science in Geodetic Engineering.** **Oct 2020 - Oct 2022**
Major: Geoinformatics
University of Bonn, Germany
Grade: Excellent (1.4)
- **Bachelor of Science in Geomatic Engineering and GIS.** **Sept 2013 - Nov 2018**
Major: GIS and Remote Sensing
Jomo Kenyatta University of Agriculture and Technology (JKUAT), Nairobi, Kenya.
Grade: Second class honours upper division

Thesis and Publications

- Kenduiywo, B.; Mendez, A.; Liebig, T.; Belli, A.; Villa, V.; Achicanoy, H.; **Korir, V.**; Chepngetich, B.; Pacillo, G.; Laderach, P. (2023) Mapping climate insecurity hotspots: Enhancing climate peace and security decision making in East Africa and greater Horn of Africa. <https://hdl.handle.net/10568/134672>
- Bridging Climate Change and Human Mobility— Collective analysis for a deeper understanding of climate mobility in the East and Horn of Africa. IOM, Global. <https://dtm.iom.int/reports/bridging-climate-change-and-human-mobility-collective-analysis-deeper-understanding-climate>
- **Korir, V.**, Forsch, A., Dehbi, Y., and Haurert, J.-H.: Visualizing the Modal Split in Public Transportation Networks, Abstr. Int. Cartogr. Assoc., 5, 89, <https://doi.org/10.5194/ica-abs-5-89-2022>, 2022.
- **Master thesis:** Land suitability analysis for placement of hydrogen infrastructure, a MCDM approach: case study: ECOWAS nations, West Africa.
- **Bachelor thesis:** Use of GIS and Remote Sensing technologies for monitoring and forecasting fall armyworm *Spodoptera frugiperda* infestation, case study: Transzoia county, Kenya.

References

- Victor Villa
 Research specialist, Alliance of Bioversity and CIAT
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- Adrian Strauch
University of Bonn, Center for Remote Sensing of Land Surfaces(ZFL)
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- Dr. Asad Asadzadeh
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